Well-Posed Learning Problem (Exam Notes)

A computer program is said to learn from **experience (E)** with respect to some **task (T)** and **performance measure (P)** if its performance at task **T**, as measured by **P**, improves with experience **E**. This definition was proposed by **Tom M. Mitchell**.

A well-posed learning problem consists of three important components:

1. Task (T):
The task refers to the specific problem that the machine learning model is trying to solve. It defines what the system is expected to do.
Example: Classifying emails as spam or not spam, predicting house prices.

2. Experience (E):
Experience refers to the data or past information used by the model to learn. The system improves its performance by learning from this data.
Example: Previously labeled emails or historical datasets.

3. Performance Measure (P):
Performance measure is used to evaluate how well the model is performing the given task. It quantifies the improvement in learning.
Example: Accuracy, Mean Squared Error.

Thus, a learning system is considered effective if its performance on a given task improves with experience. If there is no improvement in performance, then learning has not occurred.

Example:
Consider a spam email classification system:

- Task (T): Classify emails as spam or not spam
- Experience (E): Dataset of labeled emails
- Performance (P): Accuracy of classification

Hence, the concept of a well-posed learning problem helps in clearly defining and evaluating machine learning systems.